Analyzing Patterns and Predicting Severity of Traffic Accidents in Tennessee: The Impact of Weather and Temporal Factors

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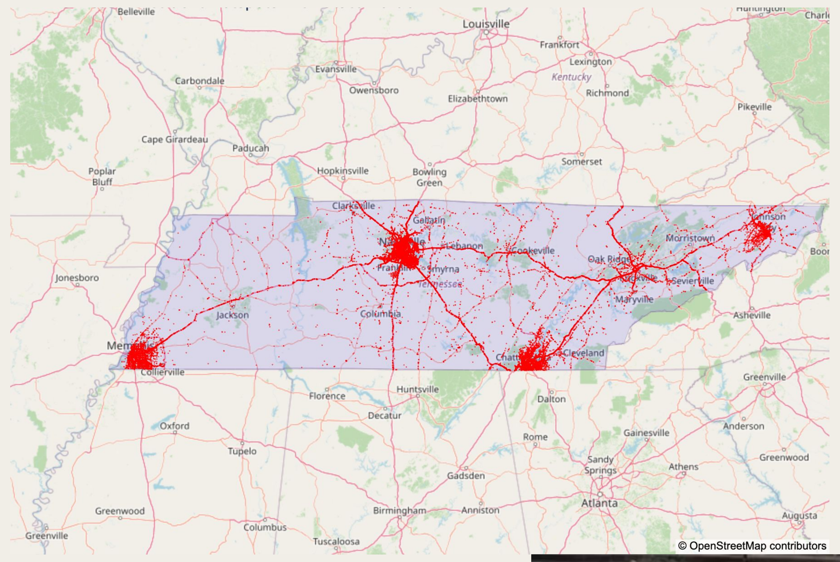
Motivation

Traffic accidents are a leading cause of fatalities in Tennessee, with weather and time factors playing critical but poorly quantified roles.

The goal of the project was to:
- Identify how weather conditions (rain, fog) and temporal patterns (rush hour, daylight) jointly influence accident frequency and severity.
- Build predictive models to prioritize high-risk scenarios for targeted interventions.

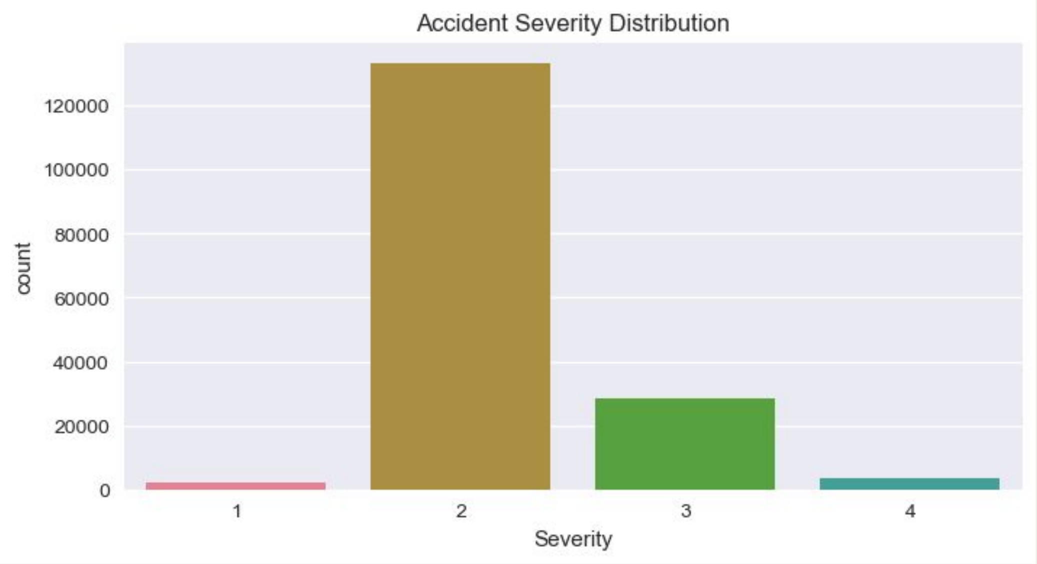


Data

Dataset: US Accidents (2016-2023) filtered to Tennessee (167,388 accidents, 46 features).

Key Features:
- Temporal: Hour, weekday, and month.
- Weather: Temperature, visibility, precipitation, weather conditions.
- Target: Severity (1-4) and derived binary severity (Low = 0: 1-2, High = 1: 3-4).

Preprocessing:
- Handled missing/irrelevant data (167K accidents, 11 Features).
- Engineered features: 'Time_of_Day' (Night/Morning/Afternoon/Evening), 'Rain_Rush' (rain during rush hour), etc.



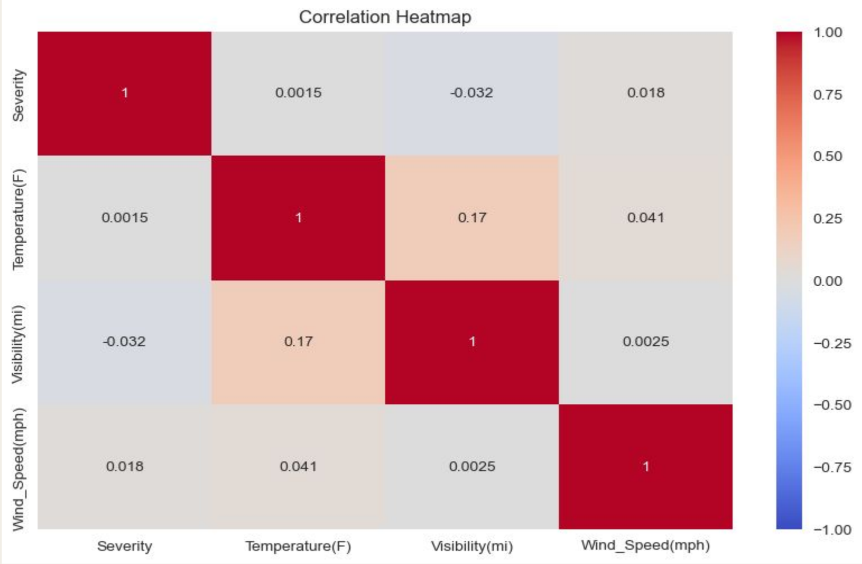
Methodology

I used Supervised Learning Algorithms. Used different methods to get results.

For Accident Frequency:
- Poisson/Negative Binomial Regression
- Predictors: Rush hour, weather type (e.g., "Light Rain" vs. "Fair")

For Severity Predictions:
- Random Forest: Binary Classification (Low/High severity)
- Ordinal Logistic Regression: Redicts original severity levels (1-4).

Tools used:
- Python (Pandas, Scikit-Learn, StatsModels)
- Seaborn/Matplotlib

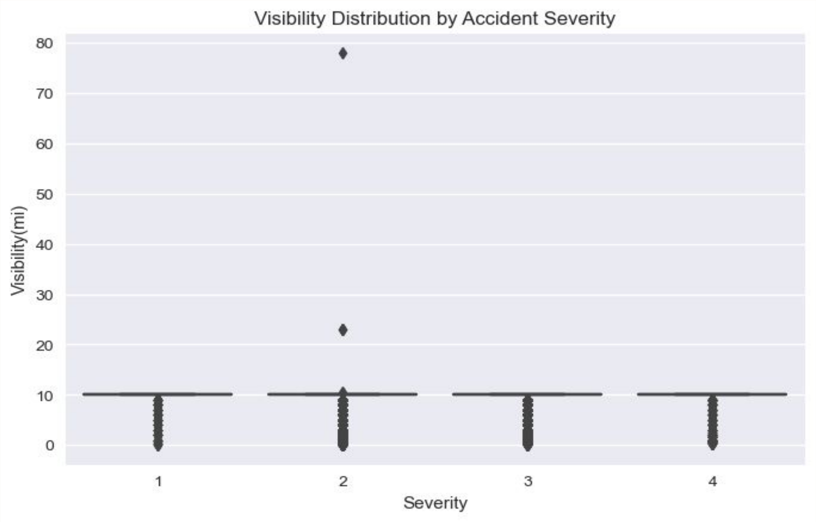


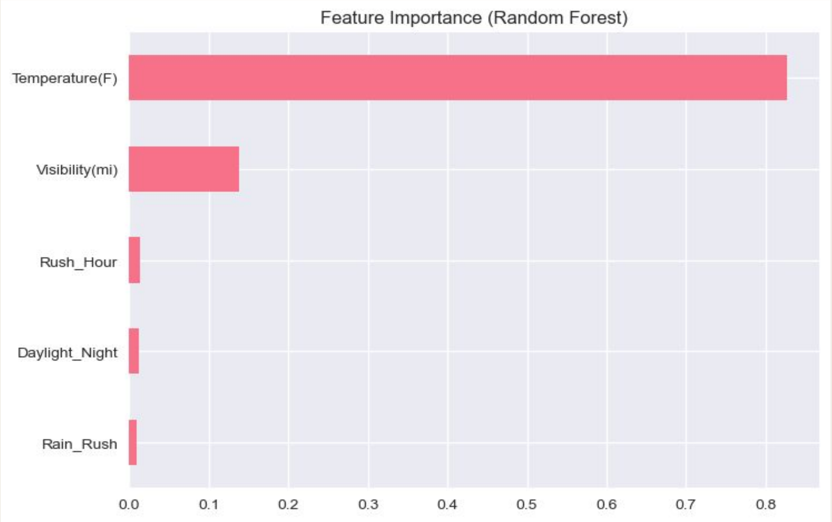
Findings

Frequency Model:
- Rush hour increases accidents by 44%
- Light rain reduces accidents by 34% vs. fair weather. The reason is most likely due to reduced traffic volume.

Severity Models:
- Random Forest: Modest discrimination (ROC-AUC = 0.61); struggles with high-severity cases (precision = 27%).
- Ordinal Logistics: Better ordinal ranking (MSE = 0.27, ~0.5 severity point error).

Key Interaction: Rain during rush hour increases severity by 2.9%.





Future Work

After spending a while trying different regression methods to improve the results, unfortunately I was not able to better it.

Feature Engineering:
- Include spatial features and traffic volume proxies.

Model Refinements:
- Address the class imbalance issue and test XGBoost for severity predictions.

Further improvements:
- Could explore other regression models with better features.
- Better data cleaning with better engineered features.

Limitations:
- Class imbalance in severity data

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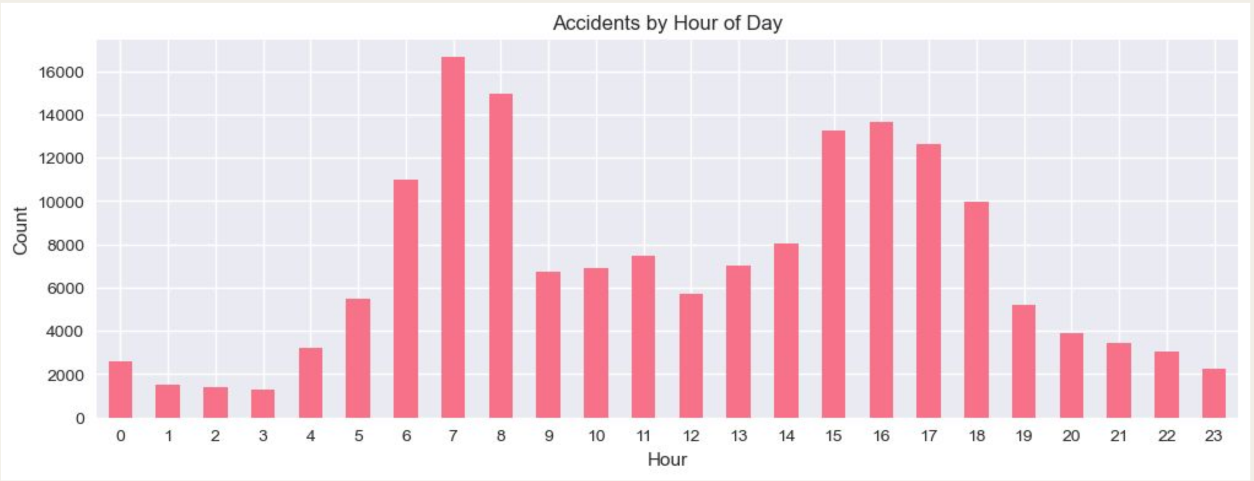
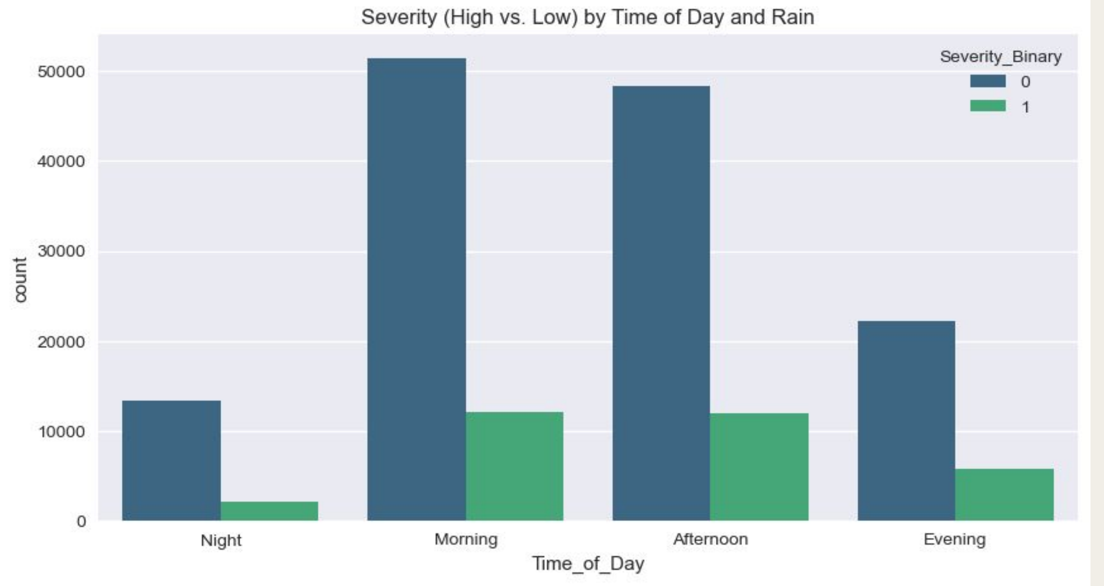
References

- Dataset: Moosavi, S. US Accidents (2016-2023), Kaggle, 2023





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